SIMATS Engineering • Department of Computer Science and Engineering • CSA0209 — C Programming • 29-08-2026 • Category: CO4_AT2_Case study • Student Submissions Report

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Q1. Memory saving

CODE

```
#include <stdio.h>

union SensorReading {
    int heartRate;
    int stepCount;
    float sleepQuality;
};

int main() {
    union SensorReading reading;
    int choice;
    scanf("%d", &choice);

    if (choice == 1) {
        scanf("%d", &reading.heartRate);
        printf("Heart Rate: %d bpm\n", reading.heartRate);
    }
    else if (choice == 2) {
        printf("Enter step count: ");
        scanf("%d", &reading.stepCount);
        printf("Step Count: %d\n", reading.stepCount);
    }
    else if (choice == 3) {
        printf("Enter sleep quality: ");
        scanf("%f", &reading.sleepQuality);
        printf("Sleep Quality: %.2f\n", reading.sleepQuality);
    }
    else {
        printf("Invalid choice\n");
        return 0;
    }

    printf("Memory used by union: %zu bytes\n", sizeof(reading));

    printf("\nUnion is preferred because only one reading is needed at a time.\n");
    printf("All members share the same memory location.\n");

    return 0;
}
```

INPUT

```
1
72
```

OUTPUT

```
Heart Rate: 72 bpm
Memory used by union: 4 bytes
```

```
Union is preferred because only one reading is needed at a time.
All members share the same memory location.
```

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Q2. Structure pointer

CODE

```
#include <stdio.h>

union ClaimDetails {
    char health[100];
    char vehicle[100];
    char property[100];
};

struct Claim {
    int claimID;
    int type;
    char policyNo[20];
    char date[20];
    char status[30];
    union ClaimDetails details;
};

int main() {
    struct Claim c;
    struct Claim *ptr = &c;

    scanf("%d", &ptr->claimID);
    scanf("%d", &ptr->type);
    scanf("%19s", ptr->policyNo);
    scanf("%19s", ptr->date);

    if (ptr->type == 1) {
        scanf("%99s", ptr->details.health);
    }
    else if (ptr->type == 2) {
        scanf("%99s", ptr->details.vehicle);
    }
    else if (ptr->type == 3) {
        scanf("%99s", ptr->details.property);
    }
    else {
        printf("Invalid claim type\n");
        return 0;
    }

    scanf("%29s", ptr->status);

    printf("--- Claim Details ---\n");
    printf("Claim ID: %d\n", ptr->claimID);
```

```
    printf("Policy No: %s\n", ptr->policyNo);
    printf("Date: %s\n", ptr->date);
    printf("Status: %s\n", ptr->status);

    if (ptr->type == 1)
        printf("Illness: %s\n", ptr->details.health);
    else if (ptr->type == 2)
        printf("Accident: %s\n", ptr->details.vehicle);
    else
        printf("Property Damage: %s\n", ptr->details.property);

    return 0;
}
```

INPUT

```
101
2
POL12345
23-08-2026
CarAccident
Pending
```

OUTPUT

```
--- Claim Details ---
Claim ID: 101
Policy No: POL12345
Date: 23-08-2026
Status: Pending
Accident: CarAccident
```

Q3. Food delivery

CODE

```
#include <stdio.h>

struct Customer {
    char name[50];
    char phone[15];
};

struct Restaurant {
    char name[50];
    char location[50];
};

struct Address {
    char houseNo[20];
    char street[50];
    char city[30];
};

union Payment {
    struct {
        char walletID[30];
    } wallet;

    struct {
        char cardNumber[20];
        char cardType[20];
    } card;

    struct {
        float amount;
    } cash;
};

struct Order {
    int orderID;
    struct Customer customer;
    struct Restaurant restaurant;
    struct Address deliveryAddress;

    int paymentType;   // 1-Wallet, 2-Card, 3-Cash
    union Payment payment;
};

int main() {
```

```
    struct Order order;

    scanf("%d", &order.orderID);
    scanf("%49s", order.customer.name);
    scanf("%14s", order.customer.phone);
    scanf("%49s", order.restaurant.name);
    scanf("%49s", order.restaurant.location);
    scanf("%19s", order.deliveryAddress.houseNo);
    scanf("%49s", order.deliveryAddress.street);
    scanf("%29s", order.deliveryAddress.city);

    scanf("%d", &order.paymentType);

    if (order.paymentType == 1) {
        scanf("%29s", order.payment.wallet.walletID);
    }
    else if (order.paymentType == 2) {
        scanf("%19s", order.payment.card.cardNumber);
        scanf("%19s", order.payment.card.cardType);
    }
    else if (order.paymentType == 3) {
        scanf("%f", &order.payment.cash.amount);
    }
    else {
        printf("Invalid payment type\n");
        return 0;
    }

    printf("--- Order Details ---\n");
    printf("Order ID: %d\n", order.orderID);
    printf("Customer: %s\n", order.customer.name);
    printf("Phone: %s\n", order.customer.phone);
    printf("Restaurant: %s\n", order.restaurant.name);
    printf("Location: %s\n", order.restaurant.location);
    printf("Address: %s, %s, %s\n",
           order.deliveryAddress.houseNo,
           order.deliveryAddress.street,
           order.deliveryAddress.city);

    if (order.paymentType == 1)
        printf("Payment: Wallet - %s\n", order.payment.wallet.walletID);
    else if (order.paymentType == 2)
        printf("Payment: Card - %s (%s)\n",
               order.payment.card.cardNumber,
               order.payment.card.cardType);
    else
        printf("Payment: Cash on Delivery - %.2f\n",
```

```
                order.payment.cash.amount);

    return 0;
}
```

INPUT

```
1001
Akshaya
9876543210
FoodCorner
Chennai
12
MainStreet
Chennai
1
WALLET123
```

OUTPUT

Q4. Digital wallet application

CODE

```
#include <stdio.h>

union CurrencyAmount {
    double standardCurrency;
    double cryptocurrency;
    int rewardPoints;
};

struct Transaction {
    int transactionID;
    char timestamp[30];
    char userID[20];

    int currencyType;   // 1-Currency, 2-Crypto, 3-Reward Points
    union CurrencyAmount amount;
};

int main() {
    struct Transaction t;

    scanf("%d", &t.transactionID);
    scanf("%29s", t.timestamp);
    scanf("%19s", t.userID);
    scanf("%d", &t.currencyType);

    if (t.currencyType == 1) {
        scanf("%lf", &t.amount.standardCurrency);
    }
    else if (t.currencyType == 2) {
        scanf("%lf", &t.amount.cryptocurrency);
    }
    else if (t.currencyType == 3) {
        scanf("%d", &t.amount.rewardPoints);
    }
    else {
        printf("Invalid currency type\n");
        return 0;
    }

    printf("--- Transaction Details ---\n");
    printf("Transaction ID: %d\n", t.transactionID);
    printf("Timestamp: %s\n", t.timestamp);
    printf("User ID: %s\n", t.userID);
```

```
    if (t.currencyType == 1)
        printf("Currency Amount: %.2f\n", t.amount.standardCurrency);
    else if (t.currencyType == 2)
        printf("Cryptocurrency Amount: %.6f\n", t.amount.cryptocurrency);
    else
        printf("Reward Points: %d\n", t.amount.rewardPoints);

    printf("Union Memory: %zu bytes\n", sizeof(t.amount));

    return 0;
}
```

INPUT

```
1001
23-08-2026_21:15
USER101
1
2500.50
```

OUTPUT

```
--- Transaction Details ---
Transaction ID: 1001
Timestamp: 23-08-2026_21:15
User ID: USER101
Currency Amount: 2500.50
Union Memory: 8 bytes
```